Unauthorised Transaction and Card Dispute Triage Service

Project plan and corpus confirmation

1. Problem statement

Banks receive a large volume of customer reports of unauthorised card and UPI transactions. A disputes analyst reads each complaint, classifies the fraud pattern, works out how many working days passed before the customer reported it, and decides the customer's liability. That decision is largely fixed by regulation: the RBI circular of 6 July 2017 on limiting customer liability sets it by *who was at fault* and *how quickly the customer reported*, and defers late reports to each bank's own Board-approved policy. The analyst then raises a chargeback, issues a provisional credit, or declines with a cited reason. Every outcome must be auditable and defensible to the RBI Ombudsman.

Doing this by hand is slow and inconsistent, because the governing rule is a two-variable table whose meaning depends on definitions ("working days", "third party breach", "customer negligence") and exceptions that sit pages away. A simple retrieve-and-answer system fails on exactly the hard cases: it returns passages that resemble the question, not the provision that modifies the answer.

What we are building. A multi-agent service that accepts a dispute report, validates it, classifies the fraud pattern, computes the reporting delay, retrieves the provisions that apply, and routes the case (chargeback initiation, provisional credit, or decline with reason) with clause-level citations. It must also know when **not** to answer: when fault is contested, provisions conflict, or the case is outside the corpus, it escalates to a human.

What it never does. It never issues a credit or closes a dispute. It produces a recommendation with evidence; a disputes analyst decides. All transaction data is synthetic; no real banking or payment system is touched.

The test case that defines success (brief section 4.1): a customer ignored a same-day SMS alert, reported three weeks later, and had earlier shared an OTP with a caller impersonating the bank. Whether that is "customer negligence" (customer bears the entire loss until reporting) or a third-party breach reported after seven working days (bank's own policy) is not settled by the circular. The correct behaviour is **escalation with both provisions cited**.

2. Objectives and success criteria

#	Objective	How we will know
O1	Reject bad input at the boundary	Every malformed request is rejected with field-level errors and quarantined; zero agent runs start for rejected input
O2	Ground every answer in the corpus	100% of returned citations resolve to retrieved text; an unverified answer is never returned
O3	Escalate instead of fabricating	Section 4.1 case and out-of-corpus cases escalate;

#	Objective	How we will know
		contested or conflicting provisions escalate with both cited
O4	Behave like a production service	Async runs behind a queue, timeouts, bounded retries, circuit breaker, graceful degradation; incomplete runs are never returned as confident answers
O5	Be observable and reconstructible	Given a run id: every tool call, query, passage, prompt version, per-step latency and cost, and verification outcome
O6	Be measured	20+ scenario evaluation set with groundedness, context relevance, faithfulness and hallucination rate, reported before and after one deliberate change
O7	Meet a declared latency target	p95 target declared before measuring (section 9), tested with Locust against a stub model

3. Scope

In scope: REST API with a published data contract; validation and quarantine; five dispute classes (card-not-present fraud, skimming, UPI social engineering, merchant dispute, duplicate charge); liability recommendation with citations; routing; escalation; web client; guardrails (prompt injection, output filtering, toxicity); evaluation harness; observability; Docker Compose deployment; runbook and incident log.

Out of scope: model training; cloud deployment; any real payment or banking integration; issuing credits or closing disputes; banks other than the one selected; languages other than English; user authentication beyond per-client rate limiting.

Assumptions: one model provider on a free tier, placed behind an interface we control; embeddings run locally; the three corpus documents are the only knowledge source.

4. Corpus confirmation

Bank selected: **State Bank of India**, because its published policy contains an explicit liability section (para 4.6) that answers the case the RBI circular defers to "the bank's Board-approved policy": beyond seven working days it states **full liability** for the customer.

Document	Issuer	Version / date	Pages	Text layer	Chunks indexed
Customer Protection: Limiting Liability of Customers in Unauthorised	Reserve Bank of	6 July 2017	9	Yes	20

Document	Issuer	Version / date	Pages	Text layer	Chunks indexed
Electronic Banking Transactions (RBI circular DBR.No.Leg.BC.78/09.07.005/2017-18)	India				
Customer Rights, Grievance Redressal and Compensation Policy 2021	State Bank of India	Version 2, 25 Oct 2021	51 (47 indexed; cover and index skipped)	Yes	154
Integrated Ombudsman Scheme, 2026	Reserve Bank of India	Effective 1 July 2026	24	Yes	78

All three have a text layer, so no OCR is needed. Source links: [add URLs]. Each file's SHA-256, page count and chunk count are recorded in the index manifest, so the index states exactly which versions it was built from.

Why this corpus is hard, and how we handled it. The liability table is meaningless without the definitions and the beyond-seven-days exception. Our chunker parses the clause structure instead of slicing by size: a clause is kept whole with its conditions; a table is one chunk together with its caption, the note after it (which carries the working-days rule and the exception) and the paragraph that introduces it; cross-references such as "paragraph 6 (ii) and paragraph 7 (ii)" are resolved into links, so a provision the query never mentions can be pulled in. Every chunk records its page and sub-clause, so citations read "para 7(i), p. 3", not just "para 7".

Items to confirm with faculty: (a) SBI's file is the Customer Rights, Grievance Redressal and Compensation Policy rather than a standalone "customer protection policy"; its liability section is para 4.6. (b) Whether a newer SBI version is expected.

5. Retrieval layer: status and evidence

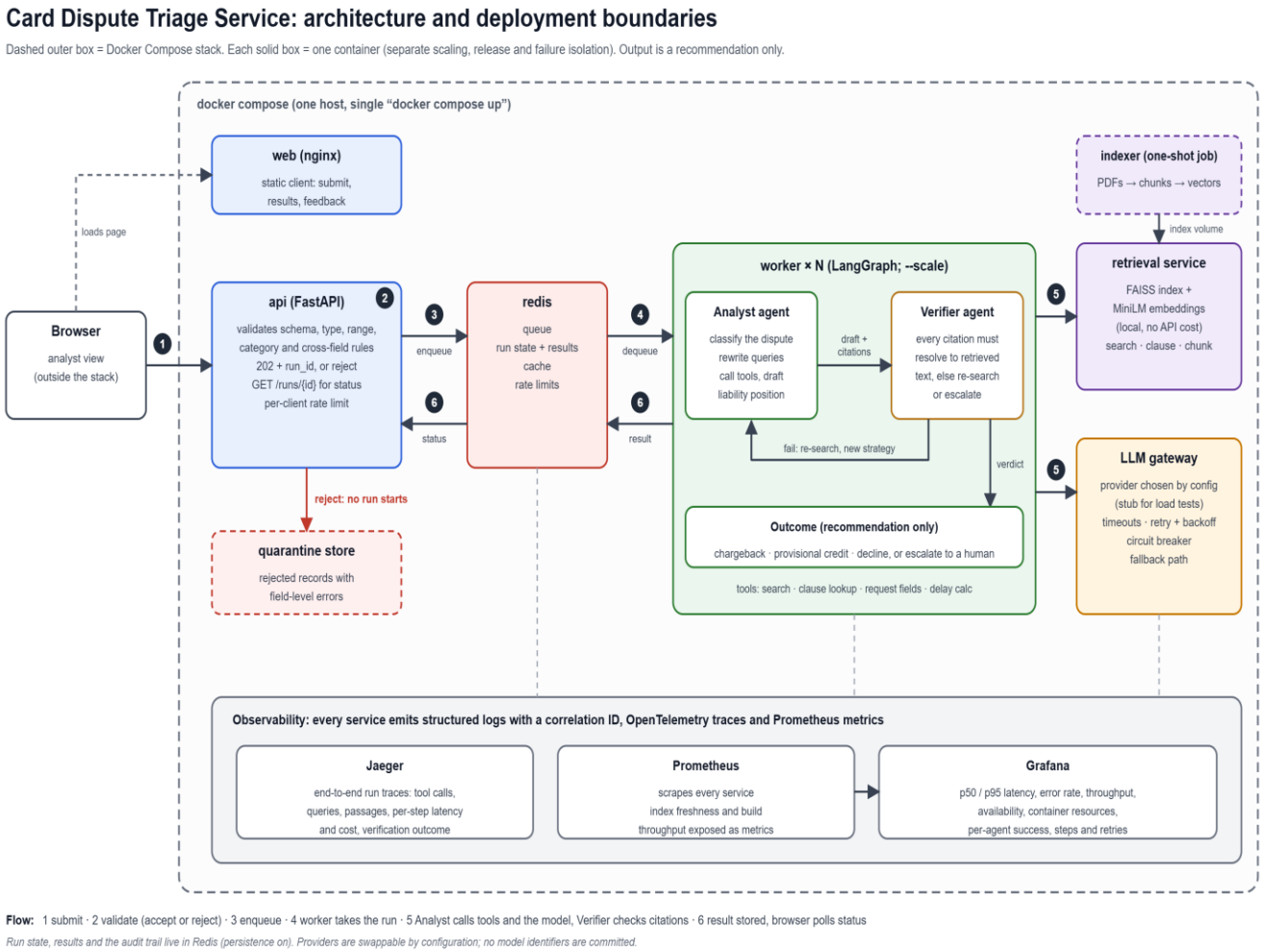
Built and tested (Member A): PDF extraction, clause-aware chunking, local embeddings (all-MiniLM-L6-v2, 512 tokens), FAISS exact index, exact clause lookup, cross-reference expansion, quote verification, an idempotent and incremental build CLI, and 32 automated tests.

Measure (10-query probe set)	Result
Expected provision in top 5	0.90
Mean reciprocal rank	0.717
Deliberate change tested: remove the document title from every chunk's embedded header	MRR 0.68 to 0.717; top-5 0.80 to 0.90 (small sample; to be re-checked on the 20-scenario

Measure (10-query probe set)	Result
	set)

Known limits that shape the design. (1) The corpus never uses the word "OTP"; it says "payment credentials". A query using "OTP" finds nothing, so the agent must rewrite queries from what it has read. (2) Similarity score cannot separate answerable from unanswerable questions: the best out-of-corpus query scored 0.535, above our weakest real hit (0.427). Escalation therefore cannot rest on a score threshold; it needs classification, a coverage check and claim verification.

6. Architecture sketch



Agent topology. Two agents with distinct responsibilities. The **Analyst** classifies the dispute, calls tools, and drafts a liability position. The **Verifier** has no authority to answer: it checks the draft against the retrieved passages only, and either approves, asks the Analyst for a different search strategy, or forces escalation. The Analyst's tools: corpus search, clause lookup by identifier, the request's own fields, and a deterministic reporting-delay calculator (working days are computed by code, not by the model).

Retrieval is a tool the agent chooses to call, not a fixed stage. The loop is bounded by steps, tool calls and wall-clock time.

If an agent fails mid-run. The run is marked failed or escalated and never returned as a confident answer. If the Analyst fails, the run retries within its budget, then escalates with the evidence gathered so far. If the Verifier fails, the draft is not released; the run escalates. Provider outage opens the circuit breaker and follows a defined fallback: a degraded result that is clearly labelled, or escalation.

Deployment boundaries (D3).

Service	Why it is separate
web (static client)	Independent release; no shared failure with the API
api (FastAPI)	Fast, stateless request handling; validation happens before any agent starts
worker (scaled with `--scale`)	Scales independently; slow runs never block the API; a crash loses one run, not the service
redis	Queue, run state, cache and rate limits in one small dependency
retrieval (FAISS + embedding model)	Different resource profile (model in memory); avoids loading the model in every worker; re-indexing releases independently
indexer (one-shot job)	Idempotent corpus build run on demand
llm-stub	Fixed responses after a realistic delay, so load tests measure our system, not the provider's rate limits
prometheus, grafana, jaeger	Observability stack; failure must not affect serving

7. Technology choices and what we rejected

Decision	Chosen	Rejected, and why
Reasoning layer	LangGraph	CrewAI: less explicit control of loop bounds and state
Run execution	Redis-backed task queue with separate workers	FastAPI background tasks: not durable, run inside the API process, cannot scale separately
Result delivery	Client polls a status endpoint	Callbacks: need an inbound endpoint on the client and more failure modes for a browser client
Vector index	FAISS flat (exact)	Chroma/approximate indexes: only 252

Decision	Chosen	Rejected, and why
		chunks, so exact search is instant and adds no recall error
Retrieval orchestration	LangChain adapter around our retriever (planned, see section 8)	Rebuilding retrieval inside LangChain: no benefit, would discard tested behaviour
Embeddings	Local sentence-transformers	Paid API: only operation with a real cost
Durable run records	Redis with persistence	Separate database: extra service for little gain at this scale
Evaluation	TruLens or equivalent, with the method for each metric documented	Ad hoc manual review: not repeatable
Load testing	Locust with a stub model	Real provider: free-tier rate limits would be what we measure

Prompts are versioned files; the prompt version is recorded with every response. No credentials, endpoints or model identifiers are committed; configuration is injected at runtime with a documented `.env.example`.

8. Requirement priorities

The brief states that completing every requirement is not expected and that prioritisation is assessed. If the system does not start and serve a valid request at the demonstration, the project scores zero, so we build a working end-to-end path first and deepen it afterwards.

Tier	Requirements	Minimum credible version
1. Spine (weeks 1 to 2, demo gate)	D3, D4, F1/F2/D1, A4, A1, L1, F3 to F6, A8	Clean-clone `docker compose up`; validated request enters a queue; worker runs a tool-calling agent over real retrieval; web client shows a cited recommendation
2. Agent and safety (week 3)	A2, A3, F7, section 4.1 case, L4, A5	Analyst plus Verifier; unverified answers blocked; escalation rules; injection and output filters; timeouts, retry, circuit breaker, fallback
3. Measure and operate (weeks 4 to 5)	L3, L2, D5, A6, D6, A7, L5, D7	20-scenario evaluation with before/after; two prompt variants compared; tracing and dashboard; Locust against the declared target; `--scale 3` effect; cost per 100 runs; caching and per-client rate

Tier	Requirements	Minimum credible version
		limiting
4. Reduced or last	L6, D8, D2 metrics	Feedback endpoint persisted and aggregated; automatic scoring of a small live sample only if time allows; runbook for two failure modes; incident log (started now)

Deliberate cuts: CrewAI and a local Ollama model are not used; callbacks are not implemented; live quality charting is the first item dropped if schedule slips.

Known gap, planned: requirement L1 asks for retrieval orchestrated with LangChain or LlamaIndex. Our retriever does not yet use either. Member A will add a thin LangChain retriever adapter so the agent tools consume it through LangChain.

The ten demonstration items and what they need

Item	Needs
1 Clean-clone compose	D3, D4
2 Valid request, cited recommendation	F5, F6, L1, A8
3 One run traced end to end	D5
4 Malformed request rejected, no run started	D1, F2
5 Corpus cannot answer, so escalate	F7, A3
6 Self-verification catches an unsupported answer	A2, A3
7 Prompt injection and guardrail response	L4
8 Circuit breaker opens, degradation path	A5
9 Locust results, failing concurrency, effect of scaling	D6, A7
10 The section 4.1 case handled correctly	F7, A3, L1

9. Declared performance target (proposed; to be frozen before any measurement)

p95 end-to-end latency, submission to final result: 30 seconds, at a sustained 6 submissions per minute with 3 workers, a stub model that answers after 2 seconds per call, and at most 6 model calls per run.

Justification (brief section 1). Disputes are worked from a queue by analysts whose own review takes minutes, so this is a triage service and not an interactive chat. A run that returns in about half a minute is well inside the analyst's workflow, and well inside our hard wall-clock cap per run (60 s). The arithmetic: 6 calls at 2 s is 12 s of model wait, leaving about 18 s for queueing, retrieval and orchestration. At 6 runs per minute, 3 workers run at roughly half utilisation, so queueing should be small. We will report requests per second, p50 and p95 against this target, state whether it was met,

and identify the concurrency at which it fails and the responsible component. This section is dated and will not be edited after the first load test.

10. Open questions

1. Is the SBI Version 2 (2021) policy acceptable as the bank document, or is a newer version expected?
2. What format and depth does Evaluation 1 expect for the plan?
3. Which model provider and free tier will the team use, and what spending limit is set?
4. Team size and final role split, and the dates of Evaluation 1 and 2.
5. Whether the proposed 30-second p95 target should be adjusted before it is frozen.